

Grounding DINO

Initial Findings

James Corino · August 2026



Why Grounding DINO?

1

Open vocabulary

Detects objects from a natural-language prompt at inference time — no fixed class list, no retraining.

2

Deep vision–language fusion

Text guides the detector throughout the pipeline, not just at a final classification step.

3

Tunable output

Box threshold filters candidate boxes; text threshold controls prompt-word matching in labels.

ROLE IN OUR PIPELINE

Stage 1 — Grounding DINO
locate faces / facial regions



Stage 2 — dedicated forensic model
classify manipulation on cropped faces

Setup: official Swin-T checkpoint · macOS /
Apple M2, CPU-only · transformers pinned to
4.37 · ~3.4 s per image

Experiment Design

21 test images



Frontal — 10

LFW (HF mirror)

4 categories



Multiple — 4

stock photos

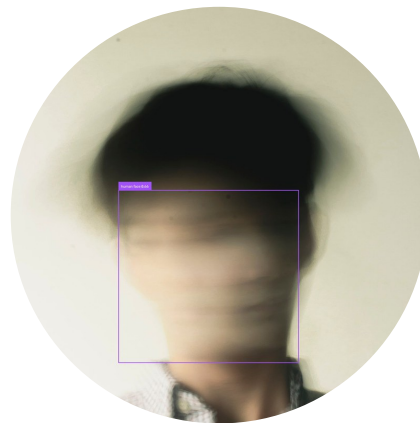
5 prompts



Profile / angled — 5

stock photos

3 threshold settings



Occluded / blurred — 2

stock photos

Prompts: "human face" · "person" · "face . eyes . mouth" · "real face" · "manipulated face" ; Batch script for saving annotated images + JSON (boxes, phrases, scores, per-image time).

Prompt Behavior

“face . eyes . mouth” multiplies boxes by design — 3–11 boxes/image vs. 1–4 for “human face”: three grounding targets, not higher sensitivity.

“real face” / “manipulated face” add no forensic signal — box counts track “human face” almost exactly (ex: [18,3,5,6] vs [17,3,5,6]); the model matches on “face”, so these prompts stay exploratory.

“person” ≠ “human face” — identical on profile/occluded (1 box each), but diverges with scene complexity: 10 vs 18 boxes on the densest crowd, 2.3 vs 1.6 on single faces.

Detection held up on profile & occluded faces — stable 1 box/image across face prompts.



“human face”
1 box

“face . eyes . mouth”
3 targets

*Same image, same thresholds ;
only the prompt changes.*

Threshold Sensitivity

No single threshold works everywhere ; raising it removes real faces, not just noise.



box = 0.25

3 boxes

box = 0.35

1 box

box = 0.45

0 boxes — face missed

Profile / occluded image across the sweep

Crowded scene: 26 → 18 → 1 boxes — 0.45
collapses a genuinely crowded scene to one face.

Frontal image: extra boxes were real background faces — not duplicates; a global cutoff discards genuine detections.

Implication: per-scene threshold policy or NMS-style post-processing needed before feeding a forensic model. Default 0.35/0.25 is a reasonable middle ground.

Limitations & Next Steps

KNOWN LIMITS

- No deepfake content tested yet — FF++ access pending
- Small categories: occluded n=2, multiple n=4
- Small / low-resolution faces not yet represented
- No universal threshold; duplicates need post-processing

NEXT

- Re-run “real/manipulated face” prompts on deepfake frames
- Expand occluded & multiple-face sets; add small/low-res condition
- Evaluate NMS / per-scene thresholds as pre-filter for the forensic classifier

Takeaway: Grounding DINO reliably localizes faces across pose and occlusion, but thresholding alone can't separate noise from real faces, and its prompts carry no indicator.

FINDINGS REPORT V2

Update — Response to Supervisor Feedback

Expanded test conditions, new failure mode, dataset-access status

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Expanded Test Set

Two conditions Tamer flagged as too thin — plus two conditions that didn't exist yet.

CATEGORY	COUNT	WHAT CHANGED
Single frontal face	10	Unchanged — LFW (HF mirror)
Profile / angled	5	Unchanged — stock photos
Multiple faces	5 +1	One new dense crowd scene (Unsplash) — see next slide
Occluded / blurred	7 +5	Masks, hands, hair, sunglasses+scarf, partial crops
Small / low-resolution	6 NEW	Thumbnail downscales (32–90px long edge) of already-curated photos, isolating resolution as the sole variable
Deepfake sample	12 NEW	Kaggle placeholder (non-forensic) — see Dataset Access below

Dataset access — FaceForensics++ / RWDF-23 / FakeAVCeleb: all three requested, **none granted yet**. Kaggle deepfake-image sample used as a clearly-labeled, non-forensic stand-in in the meantime.

A Second, More Severe Failure Mode

Not duplicate boxes this time — the model misses the scene entirely.

0 / 5

prompts detect anything usable
(default threshold)

30–40px

approx. face size in the
new stadium/rally photo

2 boxes

recovered even at the most
permissive threshold (0.25/0.20)

18 boxes

for comparison: the original
Day05 crowd outlier, same
prompt

What happened

- 4 of 5 standard prompts returned **zero detections** on the new crowd photo at default thresholds (box 0.35 / text 0.25); "manipulated face" returned 1 low-confidence box (0.44), likely noise.
- The full threshold sweep doesn't rescue it — even 0.25/0.20 recovers only 2 very-low-confidence boxes (0.30, 0.26) against hundreds of visible faces.

Why it's different

This is a different failure mode than the original crowd outlier: that image over-detects (18–26 duplicate boxes); this one is missed almost entirely once faces get small/dense enough. Not fixable by threshold tuning — a genuine resolution/density limit of the detector.

Resolution Holds Up; Prompts Still Carry No Signal

Small / low-resolution (6 images)

- **Single-subject thumbnails** (32–56px) — still 1 box each, confidence comparable to full resolution.
- **Multi-subject thumbnails** (90px) — still detect 4–5 faces, but at reduced count/confidence: partial, not complete, degradation.
- **Occluded (7 images)** — 6 of 7 land exactly 1 box; the one exception (hair covering the face) gets 2 overlapping boxes (0.60, 0.39).

Deepfake sample (12 imgs, exploratory)

PROMPT	AVG BOXES/IMG
human face	1.33
person	1.58
face . eyes . mouth	4.58
real face	1.25
manipulated face	1.17

"Real"/"manipulated" still track "human face" almost exactly — replicates the Day05 finding, now on a (non-forensic) deepfake-labeled sample. No evidence the model distinguishes genuine vs. manipulated via prompt wording alone.

Limitations & Next Steps

KNOWN LIMITS (V2)

- **Dense-crowd scenes (30–40px faces)** — zero detections, not threshold-fixable. The most significant limitation found this round.
- Still no genuine deepfake content tested — FaceForensics++ / RWDF-23 / FakeAVCeleb all pending; Kaggle placeholder in use.
- Prompt wording carries no forensic signal — confirmed again on the deepfake-labeled sample.

NEXT

- Investigate the dense-crowd failure further — e.g. tiling the image before inference, or higher input resolution.
- Swap the Kaggle placeholder for a genuine dataset once FF++/RWDF-23/FakeAVCeleb access lands.
- Evaluate NMS / per-scene thresholds as a pre-filter for the forensic classifier (carried over).

Takeaway: Grounding DINO's face-localization holds up through occlusion and down to thumbnail resolution — but breaks down entirely, not just noisily, on dense/small-scale crowds. That gap is now the primary open technical question before adding a forensic classifier stage.